Levi Skye Morris

Staff / Principal Engineer | Payments, Data & Platform Systems | Former CTO
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SUMMARY

Hands-on staff-level engineer and former CTO with 11+ years designing and building payments, healthcare and distributed systems on PostgreSQL: ledgers, payment gateways, third-party integrations and APIs that handle sensitive financial and health data. Architected and led the rebuild of a high-risk payments platform and kept it running through banking loss, deplatforming and persistent attacks, while mentoring engineers and staying in the code on the hardest problems.

EXPERIENCE

SpankChain | Remote | 2019–2026
Chief Technology Officer (Hands-on Chief Architect) | Jul 2020–Apr 2026
Acting Head of Engineering | Dec 2019–Jul 2020
Senior Software Engineer | Apr 2019–Dec 2019
High-risk payment processor for independent creators: ran its own ledger, user-to-user payments, KYC/AML and crypto settlement rails through banking loss and deplatforming.

- Architected and led the ground-up rebuild of SpankPay v2, the company's ledger and payments platform: replaced an over-normalized schema and a home-rolled framework with NestJS conventions and targeted denormalization, storing an opening balance on every transaction so balances were fast to read and auditable end to end.
- Kept balances correct under concurrent writes with serializable transactions on a single primary, planned hot-account balance splitting for scale and migrated balances via snapshot and backfill; Keycloak handled identity and access.
- Owned payment and settlement systems processing seven figures in transaction volume, including banking integrations, KYC/AML controls, GDPR/CCPA compliance and federal age- and identity-verification recordkeeping (18 U.S.C. § 2257), along with security, reliability and incident response.
- Architected and scaled the security of mission-critical payment infrastructure and cryptographic signing keys against persistent, well-resourced adversaries, and led incident response.
- Architected SpankPay.me, a payment-link product that kept independent creators paid after mainstream processors deplatformed them, and SpankMatch, defining its stack and data flow and holding final approval over design changes.
- Set technical direction across services, payments, data and blockchain infrastructure, and owned GCP operations: GKE with Helm, Cloud Run, Cloud SQL, BigQuery and Pub/Sub.
- Maintained Ethereum payment-channel infrastructure, working across signing, key management and Keccak hashing in Solidity, Vyper and JavaScript; used Python to parse complex smart-contract state from the Ethereum blockchain.
- Maintained 99.9% platform uptime through the loss of banking partners, product deplatforming and shifting regulatory requirements; retained roughly 80% of the core team through a major product sunset and restructuring.
- Grew from senior engineer to CTO, leading 8–10 engineers across five countries plus a contracted team and owning hiring, coaching and performance management; kept weekly 1-on-1s with junior and senior engineers throughout, including mentoring a fresh bootcamp hire into a highly effective IC writing production backend microservices within four months.
- Built support and escalation from zero, with a rotating triage engineer and protected capacity for technical debt, and turned roadmaps set with the CEO and CFO into epics and sprints without a dedicated project manager.
- Led the technical wind-down: kept security and access controls intact through staff reductions, archived regulated data and modeled recovery of six-figure residual balances from retired payment-channel contracts.

CampSpot | Denver, Colorado | 2017–2019
Senior Software Engineer | Sep 2017–Feb 2019

- Led a three-person payments squad; architected and built a payment gateway to PCI DSS Level 1 requirements, including third-party processor integrations, transaction workflows, failure handling and AES-256 encryption of cardholder data.
- Built and scaled customer-facing reservation and payment systems with Java, Kotlin, TypeScript, Dropwizard, Node.js, MySQL and React.

DocNetwork / CampDoc | Ann Arbor, Michigan | 2015–2017
Development Team Lead | Aug 2015–Jun 2017

- Resolved severe PostgreSQL bottlenecks through query-plan analysis, indexing, recursive CTEs and data-model changes.
- Led a 3–5-person team building a HIPAA/FERPA-compliant electronic health record and registration platform for pediatric and educational organizations.

- Ran true XP, with mandatory driver/navigator pairing on sensitive features and mob programming for the worst outages; later carried the practice to CampSpot.
- Led engineering as monthly payment volume grew from six to seven figures, balancing delivery against privacy, security, data integrity and operational health.
- Maintained an in-house flow that took raw card numbers through our own form and sent them securely to the processor.

SearchSpring | Colorado Springs, Colorado | 2014–2015

Integration Specialist | Nov 2014–Aug 2015

- Built third-party platform integrations for Shopify, Magento, NetSuite and BigCommerce, with pipelines that loaded catalog data into S3 and indexed it into Elasticsearch with custom relevance weighting; diagnosed production issues across the ELK stack.

Gelada Research Project | Simien Mountains, Ethiopia | 2012–2014

Field Site Manager | Jun 2012–Jan 2014

- Ran personnel, logistics, safety, budgets and research data collection at a remote, high-altitude primate field station under tight operational, financial and physical constraints.
- Maintained and heavily extended the station's Palm Pilot data-collection software (NSBasic) with no prior programming experience and no internet access, including a custom text-file persistence layer the platform didn't natively support.

PROJECTS

WUIXpose | 2026

Open source: Python, Django, PostGIS and React

- Building a wildfire mitigation tool that maps home ignition zones around every structure in Boulder County from parcel, footprint and address data, on Django, GeoDjango, DRF, PostGIS, React and MapLibre.
- Serves map data as PostGIS-generated vector tiles; keeps incremental jobs correct with database triggers that track geometry changes and versioned zone rules.

Wildfire risk and edge photogrammetry | 2026

Technical and business exploration

- Designed a wildfire-risk assessment service combining UAV and on-foot photogrammetry with a reporting pipeline that turns field observations into conservative, plain-language assessments, using constrained local inference checked by an independent auditing agent.

Computational ecology | 2026

Clojure, Python and scientific computing

- Built Ricker and stochastic eco-evolutionary population models in Clojure and Python (NumPy, SciPy), with equilibrium and cycle detection, Lyapunov exponent estimation and bifurcation analysis.

SKILLS

Languages: TypeScript, JavaScript, Java, Kotlin, Clojure, Python, SQL, Solidity, Vyper, R

Product and web: React, NextJS, Node.js, NestJS, Django, Ruby on Rails, Express, Dropwizard, Angular, Vue, REST, GraphQL, HTML, CSS, SASS/SCSS

Data: PostgreSQL, MySQL, MongoDB, PostGIS, Redis, Elasticsearch, GCP and S3-based data lakes

Infrastructure: AWS (S3, EC2, EKS, Lambda, RDS/Aurora, KMS, IAM, CloudFront); GCP (GKE, Cloud Run, BigQuery, Pub/Sub, Cloud SQL, IAM, Cloud Storage); Cloudflare (Workers, WAF, DDoS protection); Kubernetes, Helm, Docker, Terraform, Linux, KVM/QEMU, Proxmox, Tailscale; networking (DNS, TLS, VPC design, load balancing); Vercel

Delivery and observability: Git, GitHub Actions, Buildkite, Jenkins, CI/CD pipelines, Datadog, New Relic, Prometheus, Graylog

Process: Agile, Kanban, Scrum, XP (pairing, mob programming), Jira, Confluence, GitHub Projects, Pivotal Tracker

AI and agentic systems: multi-model agentic workflows across Claude, GPT, Gemini, Qwen and Gemma with cost- and privacy-aware routing between frontier APIs and self-hosted models (LiteLLM, Ollama)

Security and cryptography: AES-256 and field-level encryption of cardholder data and SSNs, key management, digital signatures, salted hashing (bcrypt), Keccak/SHA-family hashing, in-house authentication systems, Keycloak

Risk and compliance: PCI DSS (Level 1 build, raw card handling), HIPAA, FERPA, KYC/AML, GDPR/CCPA, 18 U.S.C. § 2257 recordkeeping; security, privacy and incident response

EDUCATION

University of Michigan | Bachelor of Science | 2012

National Science Foundation | Graduate Research Fellowship Program, Honorable Mention | 2014